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ESG performance and corporate risk-taking: Evidence from China

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ABSTRACT

Corporate environmental, social and governance (ESG) is vital for sustainable growth, while the motivation of corporate ESG engagement could decide whether ESG participation is green or greenwashing behavior. This paper attempts to understand the motivation of corporate ESG engagement from the firm's risk-taking perspective. Using Chinese publicly listed firm data from 2010 to 2020, we find that ESG rating significantly reduces corporate risk-taking. This finding still holds after a series of robustness tests to address potential endogeneity concerns and alternative risk-taking proxies. Furthermore, the marginal inhibitory impact of ESG on corporate risk-taking is more pronounced in firms with lower information transparency, weaker corporate governance and less external monitoring pressure. Our results shed essential insight on the trade-off between sustainable growth and corporate risk-taking behavior in a relatively weak investor protection institutional environment.

1. Introduction

The public and academics have an increasing awareness of corporate environmental, social and governance (ESG) performance in recent years. Firms are expected to contribute to social warfare promotion and sustainable society growth as well as their own profit maximization. Therefore, firms need to allocate certain resources to increase their ESG performance, which significantly changes the corporate strategy from solely economic focus. Corporate risk-taking levels are adjusted accordingly, as allocating recourse to ESG as a kind of insurance-like asset will reduce the corporate risk-taking level. However, if firms aim to achieve better economic performance with the privilege of ESG engagement, their risk-taking levels will be increased by chasing higher profit projects to compensate for the reduced profit of resources allocated to ESG. Therefore, the net impact of ESG performance on corporate risk-taking is unclear.

From the micro level, a certain level of risk-taking can accelerate the accumulation of corporate capital (John, Litov, & Yeung, 2008), thereby increasing a firm's core competitiveness. From the macro level, the increase in the risk-taking level among the overall level of the economy and society implies a higher level of capital and innovation expenditure (John et al., 2008; Low, 2009), which contributes to the long-term economic development (Acemoglu & Zilibotti, 1997; Baumol & Strom, 2007). Most of the previous studies on the factors influencing corporate risk-taking have explored the internal and external characteristics of the firm and further analysed the firm's risk-taking based on these characteristics, including the firm's shareholding structure (Faccio, Marchica, & Mura, 2011), CEO characteristics (Faccio, Marchica, & Mura, 2016), internal governance of the firm (Anantharaman & Lee, 2014), and legal system (Acharya, Amihud, & Litov, 2011; Bargeron, Lehn, & Zutter, 2010), cultural differences (Li, Griffin, Yue, & Zhao, 2013), etc. Since these characteristics are time-varying, firms' risk-taking preferences tend to adjust continuously with changes in internal and external enviroment (Gopalan, Gormley, & Kalda, 2021).

Dunbar, Li, & Shi (2020) argue that ESG engagement promotes a firm's ability to afford risks in the future. Firms take higher levels of compensation incentive contracts with CEOs after ESG engagement to encourage CEOs to take risks and enhance CEOs' risk-taking motivation. However, a CEO's risk-taking motivation does not imply an increase in the actual level of a firm's risk-taking, and the impact of ESG on firms is not solely rely on the CEO's risk-taking motivation. There is little empirical evidence that ESG engagement could induce higher compensation incentive contracts in the Chinese capital market. On the one hand, the external trust and operating cost advantages brought by ESG

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engagement make firms have higher risk-taking ability in the future, and the risk preference of firms might be more aggressive in achieving the goal of maximizing future value and shareholders' wealth under the protection of ESG. On the other hand, becoming a socially responsible firm and gaining a good reputation normally set high standards for firms in terms of ethical, legal and public relationship. Managers might be more conservative in taking risks to prevent the loss of social capital and reputation. Therefore, the impact of ESG performance on corporate risk-taking need to be tested.

An increasing number of companies have begun to integrate ESG into all aspects of their daily business operations. The awareness of ESG in China is increasing year by year. In 2008, the Shanghai and Shenzhen exchanges required certain firms to disclose corporate social responsibility reports along with their annual reports and encouraged other listed companies to voluntarily disclose social responsibility reports. The number of firms that disclosed their ESG information in China has grown from 371 in 2009 to 1092 in 2020, and the importance of ESG engagement has become increasingly prominent. However, firms that have engaged in ESG activities are still plagued by various negative events, such as vaccine maker Changsheng Bio, which donated 7.4 million worth of drugs in the Wenchuan earthquake and caught up in the recent fake vaccine scandal. The managers of these firms used social responsibility as a self-interest tool to cover up their own misconduct behavior and pursue personal reputation, wasting corporate resources to maximize their self-interest and exposing the companies to greater risks.

As the main economic foundation of the economy and society, firms are the main creators of social wealth. A proper level of risk-taking is beneficial to the development of the firm (John et al., 2008). While firms might invest in ESG for safety reasons to protect theirselves in the case of negative events (Godfrey, Merrill, & Hansen, 2009), there is also motivation for firms to achieve impact investing with higher profit. These different motivations will lead to different outcomes in corporate risk-taking performance. We argue that higher ESG performance could decrease corporate risk-taking, as the overload in ESG activities crowds out firms' business resources and makes managers more prudent in business decisions.

First, from an external monitoring perspective, ESG attracts customer interests and improves a firm's financial performance (Servaes & Tamayo, 2013). Firms with better ESG performance attract more external resources (Lins, Servaes, & Tamayo, 2017) and gain more government trust. ESG engagement is an investment in social capital that provides firms with an insurance-like effect against poor market performance in the case of negative news (Godfrey et al., 2009; Jo & Na, 2012; Oikonomou, Brooks, & Pavelin, 2012). Being socially responsible and becoming a star firm in ESG engagement could gain considerable reputation and public attention, which means that firms are more transparent to their stakeholders. The information disclosed in ESG reports will be disseminated by the media (Groza, Pronschinske, & Walker, 2011), and firms are subject to stronger external monitoring from the public (Porter & Kramer, 2006). This external pressure on managers will make them focus more on their career reputation and avoid investing in risky projects. The better ESG performance is, the less risk managers will take to prevent reputation loss.

Second, from an agency cost perspective, firm managers are generally considered to be risk averse in modern corporate governance theory due to the presence of external reputation and career risk (Akbar, Kharabsheh, Poletti-Hughes, & Shah, 2017; Faccio et al., 2016). Companies are socially responsible for cover-up motives, which aim to reduce the cost of covering up negative information and increase the difficulty for investors to discover the negative information. Managers tend to choose ESG investment to avoid high-risk investment to satisfy their self-interest and future career plans. Due to limited resources, the opportunistic behavior of managers to overcommit to social responsibility to build their good reputation makes it difficult to cover the costs of ESG engagement for the firm (Borghesi, Houston, & Naranjo, 2014). Therefore, managers has the opportunistic motivation of overinvesting in ESG to enhance their personal reputation and lay a good foundation for their future careers at the expense of stakeholders.

To explore the impact of ESG performance on corporate risk-taking, we use the comprehensive ESG rating of Chinese publicly listed firms published by Hexun.com (He, Du, & Yu, 2022; Hao & He, 2022) as an explanatory variable to examine its impact on corporate risk-taking, which is proxied by a joint indicator of corporate cash holdings and net leverage (Bernile, Bhagwat, & Rau, 2017; Faccio et al., 2011; Gopalan et al., 2021). A higher level of firm cash holdings with a lower level of leverage imply a lower level of corporate risk-taking. Our empirical results show that ESG rating reduces the level of corporate risk-taking. To further verify the reliability of the results, a series of robustness tests are conducted to avoid potential endogeneity issues, including controlling firm fixed effects for omitted variables, alternative explanatory variables for measurement bias, PSM regression for sample selection bias, and an instrumental variables regression. Our results still hold after these robustness checks.

Furthermore, we test the potential channels through which ESG affects corporate risk-taking. We find that better ESG performance will lead to higher firm transparency, lower managers' excessive perks and less negative news, which further increases corporate risk-taking. These channels are further supported by our cross-sectional heterogeneity tests. The marginal impact of ESG performance on corporate risk-taking is more pronounced in firms with higher information asymmetry, lower corporate governance and lower reputation. Therefore, ESG performance could affect corporate risk-taking through its nonfinancial information supplements effect, corporate governance effect and reputation-building effect.

We contribute to the literature in the following ways. First, this paper empirically investigates the relationship between corporate comprehensive ESG performance and the level of corporate risk-taking. Complementing previous literature that focuses on internal and external characteristics of firms to analyse changes in risk taking (Faccio et al., 2016; Boubakri, Cosset, & Saffar, 2013; Acharya et al., 2011), we focus on the firm's comprehensive ESG performance and find a new influencing factor of corporate risk-taking.

In addition, complementary to the existing studies of the impact of ESG on financial performance (Hong, Li, & Minor, 2016; Godfrey, Hatch, & Hansen, 2005; Orlitzky & Benjamin, 2001; Lins et al., 2017), this paper empirically demonstrates that corporate risk preference is affected by ESG through three channels. We provide empirical evidence on the economic consequences of ESG from the perspective of risk-taking and provide a policy basis for further development of the positive impact of ESG.

Finally, our research findings in China, as an important emerging economy, could bring general implications to understanding the effectiveness of ESG in a relatively weak corporate governance market, which has significant insights for investors to understand the economic consequences of ESG in China.

The remainder of the paper is organized as follows: section 2 reviews the relevant literature and proposes our hypotheses; section 3 presents the data, variables and research design; section 4 presents the empirical results, including the main regression results and a series of robustness tests; section 5 tests the possible channels and cross-sectional heterogeneity; and we conclude in section 6.

2. Literature review and hypothesis development

2.1. Relevant literature

Risk-taking refers to the pursuit of risky assets in business activities with the aim to achieve higher corporate value in the long run and maximizing shareholders' interests (Daron & Zilibotti, 1997). Numerous studies have explored various factors that influence corporate risk-taking from different perspectives. A firm's risk-taking level is the degree to which it could take and control risk; It reflects the firm managers'

attitude and willingness to make risky decisions under an uncertain business environment or the degree of their risk aversion, which mainly decided by the strategy choice of managers. The factors that influence corporate risk-taking are explored mainly through three aspects: the external corporate environment, the corporate governance level, and managerial characteristics.

From the external environment of firms, the positive market feedback of firms increases their risk-taking (Habib & Hasan, 2017). John et al. (2008) argue that countries with better investor protection mechanisms encourage shareholders to invest in risky projects with NPVs greater than 0, increasing the risk-taking level of firms. Meanwhile, countries with better creditor protection environments are more assured of smooth cash flows and lead to lower levels of risk-taking (Acharya et al., 2011). Bargeron et al. (2010) mention that the enactment of the Sarbanes-Oxley Act provided penalties ranging from minor fines to severe jail for noncompliant managers, which implies that firms listed in the U.S. must ensure that not only their financial statements are accurate but also that their internal control systems can pass the relevant audits. As a result, firms will be more conservative in risk-taking.

From the corporate governance perspective, CEO heterogeneity (Quigley, Hambrick, & Misangyi, 2019), the size of the board of directors (Wang, 2012), the concentration of power in family firms (Su & Lee, 2013), and the diversity of nationality of the firm's shareholders (Boubakri et al., 2013) affect the level of corporate risk-taking. Faccio et al. (2011) find that the more diversified a firm's equity structure is, the higher the firm's risk-taking level, while later studies find that the proportion of foreign ownership further affects the firm's risk-taking level (Boubakri et al., 2013). In addition, equity incentive policies may also induce managers to change their risk attitude to be more aggressive and consistent with the goal of maximizing shareholders' interests, resulting in higher levels of risk-taking (Anantharaman & Lee, 2014; Kim, Patro, & Pereira, 2017; Lim & McCann, 2014). Santosh, Andrew, Suman, & Chandra (2018) further find that corporate governance promotion will reduce the information asymmetry between firms and investors, increasing the liquidity of the firm's stock, attracting foreign investors and enhancing the level of risk-taking.

From the firm management perspective, the main focus on the impact of managers' characteristics on corporate risk-taking is from upper echelons theory, such as managers' gender (Faccio et al., 2016; Sila, Gonzalez, & Hagendorff, 2016), experience (Carpenter, Pollock, & Leary, 2003), age (Coles, Daniel, & Naveen, 2006), religious beliefs (Jiang, Jiang, Kim, 2015), and psychological characteristics of managers such as mental accounts and overconfidence (Baker & Wurgler, 2013; Kahneman & Tversky, 1979; Li & Tang, 2010). Eisenmann (2002) suggests that when there is a good external evaluation system, managers may be conservative in their risk attitude, either because of the need to protect their reputation or because of concerns about their future reputation, and therefore, the level of risk-taking may be reduced.

2.2. Hypothesis development

Firm managers are generally considered to be risk averse in modern corporate governance theory due to the presence of external reputation and career risk (Akbar et al., 2017; Faccio et al., 2016). Faced with a high level of such risk, managers tend to consider the investment choices that are best for their self-interest and future career plans. MacCrimmon and Wehrung (1990) conduct a factor analysis based on the personal, financial and professional characteristics of managers and show that the most successful managers tend to be risk takers and good at seizing opportunities, while most mature managers tend to be risk averse and are more conservative in their risk preference. Many studies explore the factors that affect risk-taking from the perspective of reducing agency costs. A reasonable and effective managers' equity incentive and compensation system can effectively deter managers' risk aversion and moral hazard behaviors, encouraging them to make investment choices that maximize firm value (Anantharaman & Lee, 2014; Phan & Hegde, 2013).

The literature argues that ESG engagement could enhance the corporate risk-taking level, as it could create more external financing advantages and operational advantages (Ioannou & Serafeim, 2015; Godfrey et al., 2009) and thus lead to a higher-risk-bearing capacity. These conclusions are mainly drawn from developed markets. In contrast to managers, shareholders can reduce their own investment risk by diversifying their shareholdings, and they expect the firm's risk-taking level to be maintained at a relatively high level for the firm's value maximization objectives, which being conflicts with management's conservative investment strategy (Faccio et al., 2016). Therefore, the level of corporate risk-taking often depends on how the firm mitigates such agency conflict. In the context of the Chinese ESG system, ESG information does not reduce the information asymmetry between shareholders and management (Kim et al., 2017), but rather exacerbating the agency problem. In the Chinese financial market with relatively weak corporate governance, firm managers may have higher incentives and opportunities to satisfy their own interest with ESG engagement. Managers tend to use ESG to cover up negative events or diversify risk, i.e., firms tend to engage in ESG activities when they face higher litigation risk.

Corporate risk-taking is determined by the risky decisions of managers, which are picked from feasible investment projects that combine various factors. Therefore, it is important to examine the motivation of managers' behavior in exploring the impact of ESG on the level of corporate risk-taking. Borghesi et al. (2014) find that firms with higher shareholdings of institutional investors tend to not invest too much in ESG, which questions the claim that ESG is aimed at enhancing corporate value. Ho, Kim, & Reza (2022) further find that the compensation level of CEOs is related to their reputation instead of their actual ability; therefore, the management reputation brought by ESG performance could further help managers obtain higher compensation and better future career development. ESG mainly promotes managers' external reputation and personal influence instead of firm value. Managers care more for their own interest and tend to be more conservative and prudent in risky decisions (Hirshleifer & Thakor, 1992) to maintain the operational advantages, financing advantages, and good reputation brought by the social capital of ESG. As a result, the corresponding risk-taking level of the firm will be lower. Meanwhile, the managers of the firm will also adopt a prudent investment strategy to avoid risk for their future career development (Eisenmann, 2002), which satisfies managers' own interest at the expense of the future development of the firm.

In addition, engagement in ESG is aimed at social welfare and sustainable growth in the long term. Therefore, the capital allocated to ESG activities will crowd out other business activities, which decreases corporate risk-taking in the short term.

Based on the above analysis, we propose hypothesis H1:

H1. ESG rating performance decreases the risk-taking level of the firm.

2.3. Mechanism analysis

2.3.1. Information transparency

ESG engagement and information disclosure could provide additional information besides financial information, thereby reducing firm information asymmetry. Li (2011) finds that financing constraint is an important factor affecting firms' capitalization expenditures. Firms with a higher level of financing constraint cannot bear the cash flow risk. Moreover, for firms with lower levels of financing constraints, managers have greater degree of freedom of investment choices. As a result, they tend to choose investment options that are beneficial to the long-term value promotion of the firm and behave more aggressively, which leads to a higher level of corporate risk-taking (Almeida & Campello, 2007; Fazzari & Athey, 1987; Guariglia, 2008; Houston & James, 2001). For firms with lower information transparency, it is more likely for managers to take extra risk for their self-interest. The ESG information

will be more ambiguous for stakeholders and to have a better understanding of corporate operation conditions. Previous studies have concluded that firms audited by Big 4 auditing firms or firms with less financial constraints are more transparent. Therefore, the marginal contribution of the incremental information complementary effect of ESG is expected to be weaker in these firms.

Based on the discussion above, we propose the following hypothesis:

H2. The impact of ESG performance on corporate risk-taking is more pronounced in firms with weaker information transparency, such as firms with higher financial constraints or firms that are audited by non-Big-4 auditing firms.

2.3.2. Corporate governance

A higher level of corporate governance indicates better investor protection mechanisms and lower agency costs, which will encourage managers to participate in ESG activities for stakeholder benefit rather than their own interest. A higher proportion of institutional investors' investments implies that the firm's decisions are more likely to be consistent with the goal of maximizing the firm's future value. Therefore, lower agency costs induce a higher firm risk-taking level (Eling & Marek, 2014; Jiraporn, Chatjuthamard, Tong, & Kim, 2015; John et al., 2008), which is beneficial for firms' long-term development. In firms with higher agency costs, managers are more likely to participate in ESG for their own reputation and future careers. Therefore, the impact of ESG on corporate risk-taking will be more pronounced for firms with higher agency costs.

Based on the discussion above, we propose the following hypothesis:

H3. The impact of ESG is more pronounced in firms with higher agency costs, such as in firms with a lower portion of institutional investors or in firms with higher manager perks.

2.3.3. External monitor

Managers make their risk choices based on various factors, which in turn affect the risk-taking level of the firm. For example, overconfident managers tend to be more aggressive in their risk choices (Li & Tang, 2010), but overly aggressive or conservative risk choices by managers are detrimental to the firm. It requires more effective internal and external monitoring to regulate and correct managers' self-interest or incorrect behavior. When firms face higher levels of external monitoring, such as analyst coverage and research attention, it will help to monitor managers to make correct decisions. When firms face higher levels of external monitoring, financial information is more transparent, and agency costs are reduced (Suharyono, 2019; Luo, Wei, & He, 2023).

Based on the discussion above, we propose the following hypothesis:

H4. The impact of ESG is more pronounced in firms with stronger external monitoring pressure, such as in firms with higher analyst coverage or in firms with more analyst reports.

3. Data, variables and research design

We select the data of Chinese listed firms from 2010 to 2020 and finally obtain 26,427 observations after excluding the financial industry, special treatment firms, and firms with missing data for the main variables. The ESG ratings data in this paper are obtained from Hexun.com, while other firm-level data are mainly from the CSMAR and CNRDS databases. The continuous financial data are winsorized at the 1% and 99% levels.

Based on previous research, we use a firm's cash holding level (*Cash*) and a firm's net leverage ratio (*Net Lev*) as measures of a firm's risk-taking level (Bernile et al., 2017; Faccio et al., 2011; Gopalan et al., 2021). The cash holding level (*Cash*) is the total cash and cash equivalents at the end of each year divided by the firm's total assets in the same period, and the net leverage ratio (*Net Lev*) is the normalized value of the firm's total liabilities minus cash and cash equivalents at the end of the year divided by the firm's total assets. The lower the cash holdings and the higher the net leverage are, the higher the level of corporate risk taking. In addition, we also use some alternative indicators to measure corporate risk-taking: (i) Merton default distance; the smaller the value of the firm's Merton default distance, the higher the firm's default probability and the higher the corporate risk-taking; (ii) the volatility of ROA from period t to period t + 5 adjusted by industry average (Faccio et al., 2011). The higher volatility of the firm's future ROA indicates a higher level of risk-taking; (iii) the maximum deviation of the firm's ROA over the next five years; (iv) the volatility of the annualized return on the firm's stock in period (t-2, t + 2) (John et al., 2008). These four variables are used for robustness checks for the risk-taking level of the firms.

$$Risk_i = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N}\left(Adj_Roa_{i,n} - \frac{1}{N}\sum_{n=1}^{N}Adj_Roa_{i,n}\right)^2} \Big/ \mathrm{N} = 5 \tag{1}$$

$$Adj_{Roa_{i,n}} = \frac{EBITDA_{i,n}}{Assets_{i,n}} - \frac{1}{X_n}\sum_{k=1}^{X}\frac{EBITDA_{k,n}}{Assets_{k,n}} \tag{2}$$

$$Risk2_i = Max\left(Adj_{Roa_{i,n}}\right) - Min\left(Adj_{Roa_{i,n}}\right)\Big|_{N=5} \tag{3}$$

$$Risk3_i = \sqrt{\frac{1}{N-1}\sum_{n=1}^{N}\left(Adj_Return_{i,n} - \frac{1}{N}\sum_{n=1}^{N}Adj_Return_{i,n}\right)^2} \tag{4}$$

$Risk_i$is the volatility calculated by the rolling window of the ROA from year t + 1 to t + 5 as eq. (1), where *Adj_Roa* is the mean of ROA adjusted by the industry as shown in eq. (2) (Faccio et al., 2011). $Risk2_i$ is the maximum difference value of the industry mean-adjusted ROA from period t + 1 to t + 5 (Faccio et al., 2011), which is calculated by eq. (3); $Risk3_i$ is the volatility of the industry mean-adjusted annualized return on the firm's stock over a five-year period (Coles et al., 2006; John et al., 2008), which is calculated by equation (4). These three indicators are calculated with a 5-year time window; therefore, the sample size decreases to 10,795 with these three indicators.

The ESG score published by Hexun.com, which evaluates the status of corporate social responsibility performance, is applied as the main explanatory variable. The ESG score is calculated from five perspectives, shareholder responsibility, employee responsibility, supplier, customer and consumer responsibility, environmental responsibility and social responsibility, to comprehensively evaluate a firm's ESG performance. The higher the ESG score is, the better the firm's social responsibility performance.

Following John et al. (2008), Gopalan et al. (2021), and Faccio et al. (2011), we select control variables to control for firm financial indicators and manager characteristics firm size (Size), Board size (Bsize), ownership structure (Ownership), firm age (Firm age), Return on assets (ROA), managers' salary (Compensation), independent directors (Indratio), and tangible assets ratio (Tang). In addition to the variables above, the cross-sectional heterogeneity test also involves variables indicating whether firm auditors are from the Big 4 auditing firms, firm financial constraints, institutional ownership, manager perks, analyst following and analyst reports. The detailed variable definitions are described in Table 1 below.

The baseline regression model is designed as follows in eq. (5).

$$RiskTaking_{i,t} = \beta_0 + \beta_1 ESG_{i,t} + \beta_2 Controls_{i,t} + \varepsilon_{i,t} \tag{5}$$

where $RiskTaking_{i,\ t}$ is measured by *Cash* and *Net Lev*, $ESG_{i,\ t}$ represents Huxun's comprehensive ESG score, and we further control for year and industry effects in the regression. The coefficient of interest is β_1, and a positive β_1 for *Cash* together with a negative β_1 for *Net Lev* indicate that ESG rating reduces corporate risk-taking.

Table 1
Variable definitions.

Type	Variables	Sign	Definition
Dependent variable	Cash ratio	Cash	Cash and cash equivalent/total asset at the end of a given year
	Leverage	Net Lev	Total liability- (cash + cash equivalent)/Total asset at the end of a given year
Alternative dependent variable	Adjusted volatility of ROA	Risk	See the Eq.2
	Adjusted Range of ROA	Risk2	See the Eq.4
	Adjusted volatility of annual stock yield	Risk3	See the Eq.5
Explanatory variables	Merton default distance	DD Merton	Distance of the firm from the default threshold based on Merton's model
	CSR score	ESG	Natural logarithm of CSR scores released by Hexun.com
Control variables	Size	Size	The natural logarithm of firm total asset
	Board size	Bsize	Natural logarithm of (1 + number of board members)
	Ownership Concentration	Ownership	Percentage of shareholding of the largest shareholder
	Firm age	Firmage	Years since the setup of a given firm
	Return on assets	ROA	EBITDA for the year divided by total assets at the end of the period
	Management Compensation	Compensation	Natural logarithm of management compensation
	Ratio of independent directors	Indratio	Ratio of independent directors to total number of directors
	Ratio of tangible assets	Tang	Ratio of tangible assets to total assets at the end of the period
Other variables	Shareholding of institutional investors	INI	Defined as 1 if the institutional investor shareholding of firm *i* exceeds the industry median for the same year, 0 otherwise
	Analyst attention	Analyst	Defined as 1 if the analyst attention of firm *i* exceeds the industry median for the same year, 0 otherwise
	Research report attention	Report	Defined as 1 if the number of research report of firm *i* exceeds the industry median for the same year, 0 otherwise
	Financial constraints	WW	Defined as 1 if the financial constraints of firm i exceeds the industry median for the same year, 0 otherwise

4. Empirical results

4.1. Descriptive statistics

Table 2 provides the descriptive statistics of the main variables in this paper. The maximum and minimum values of cash holdings (*Cash*) are 0.0191 and 0.594, respectively, with a standard deviation of 0.134, and the net leverage ratio (*Net Lev*) after normalization takes values between 1 and 0, respectively, with a standard deviation of 0.236, which shows that there are significant differences between firms' risk-taking levels. The maximum and minimum values of the independent variable ESG rating (*ESG*) are 1.585 and 4.275, respectively, showing significant differences among firms. The distribution of the other control variables is also similar to that in previous studies.

4.2. Baseline regression

Table 3 shows the empirical results of the baseline regression for the research hypotheses of this paper, where the dependent variable in Columns (1) and (3) is *Cash*, while the dependent variable is *Net Lev* in Columns (2) and (4). Columns (1) and (2) are the univariate regression results, and we further control for other firm level characteristics in columns (3) and (4). Year and industry fixed effects are controlled in all columns. The coefficients of ESG in columns (1) and (3) are both significantly positive at the 1% significance level, indicating that a higher ESG score increases the cash holdings of firms, and the estimated coefficient of ESG is significantly negative at the 1% level, indicating that firms with higher ESG scores have lower levels of net leverage. This regression result verifies hypothesis H1 in this paper; that is, the better the ESG performance is, the higher the cash holdings and the lower the level of net leverage, which indicates a lower level of corporate risk-taking.

In terms of the control variables, the results of the regression coefficients are also consistent with previous research findings; firms with longer establishment years and more assets have a higher risk tolerance and more aggressive risk attitude, which results in higher levels of risk-taking. Firms with higher ROA, a higher proportion of tangible assets, and a larger board of directors are more conservative in their risk preference and lead to a lower level of risk-taking (Faccio et al., 2016).

4.3. Endogeneity concern

Our research results may be biased due to potential endogeneity issues. First, our control variables could not include all factors, and we adopted firm-fixed estimation to address firm-level omitted variables. Second, our conclusion may be caused by the firms selection bias; therefore, we further match the firms with their characteristics by the PSM method and then perform regression based on the matching sample. Third, to formally address the endogeneity caused by reverse causality and other factors, we use 2SLS regression with the lagged ESG score. Although this instrument variable is not perfect, it satisfied the instrument variable requirement and passed the statistical tests. Finally, we calculate risk-taking with other methods to further verify the robustness of proxy selection.

4.3.1. Omitted variables

To address the potential endogeneity issue caused by omitted variables, we further control for the firm fixed effect in addition to the baseline regression that controls for industry and year fixed effects. The results are presented in Table 4. After controlling for the firm fixed effect, ESG performance is still positively correlated with cash holdings at the 5% significance level and negatively correlated with net leverage at the 1% significance level. The regression result after controlling for firm-fixed effects is consistent with the baseline regression.

4.3.2. Sample selection bias

In this paper, we use the PSM (propensity score matching) method to control for endogeneity problems caused by sample selection bias. Firms above the 70th percentile of ESG scores in each year are considered as the treatment group, and the other firms belong to the control group. We use one-to-one nearest neighbor matching with the control variables as the characteristic variables to match the treatment firms with the control group, and then we re-estimate the baseline regression with the matched firms. The empirical results are presented in Table 5. The logit regression

Table 2
Descriptive statistics.

variable	N	mean	sd	min	p25	p50	p75	max
Cash	26,427	0.174	0.134	0.0191	0.0776	0.133	0.228	0.594
Net Lev	26,427	0.577	0.236	0	0.423	0.595	0.753	1
Risk	10,795	0.0433	0.0452	0.00450	0.0169	0.0335	0.0493	0.355
Risk2	10,795	0.106	0.108	0.0110	0.0415	0.0824	0.121	0.842
Risk3	10,795	0.376	0.230	0.0777	0.220	0.318	0.466	1.397
DD Merton	26,395	1.981	0.492	−0.639	1.656	1.984	2.305	9.694
ESG	26,427	3.115	0.507	1.585	2.878	3.123	3.326	4.257
Bsize	26,427	2.131	0.192	1.609	1.946	2.197	2.197	2.565
Size	26,427	22.16	1.272	20.11	21.21	21.97	22.90	25.60
ROA	26,427	0.0498	0.0394	−0.0175	0.0202	0.0422	0.0713	0.167
Ownership	26,427	0.352	0.147	0.105	0.235	0.334	0.452	0.699
Firmage	26,427	17.01	5.957	0	13	17	21	62
Compensation	26,427	14.90	0.765	13.26	14.39	14.88	15.39	16.71
Indratio	26,427	0.374	0.0492	0.333	0.333	0.333	0.429	0.500
Tang	26,427	0.928	0.0833	0.609	0.916	0.957	0.979	1.000
Excess Perks	24,306	0.0273	2.098	−317.2	0.0147	0.0298	0.0502	73.69
Opaque	16,343	0.221	0.435	0.00750	0.0970	0.154	0.248	22.41

Note: This table reports descriptive statistics for the key variables used in our analysis. All the variable definitions are presented in Table 1.

Table 3
Baseline regression.

	(1)	(2)	(3)	(4)
	Cash	Net Lev	Cash	Net Lev
ESG	0.0319***	−0.0580***	0.0138***	−0.0324***
	(21.18)	(−20.45)	(8.31)	(−11.60)
Bsize			0.0157***	−0.0121*
			(3.32)	(−1.70)
Size			−0.0282***	0.0887***
			(−36.29)	(78.07)
Roa			0.6983***	−1.7653***
			(28.24)	(−47.52)
Ownership			0.0398***	−0.0511***
			(7.54)	(−6.50)
Firmage			−0.0013***	0.0031***
			(−8.99)	(14.66)
Indratio			0.0691***	−0.0831***
			(3.94)	(−3.17)
Tang			0.2161***	−0.0869***
			(29.48)	(−6.87)
Compensation			0.0112***	−0.0132***
			(9.27)	(−7.28)
Constant	0.1228***	0.7061***	0.3233***	−0.8780***
	(13.85)	(46.24)	(15.17)	(−26.74)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	26,427	26,427	26,427	26,427
Adj-R2	0.124	0.161	0.250	0.456

Note: This table presents the OLS regression results of ESG performance on corporate risk-taking. Columns (1) and (2) are regression results without the inclusion of control variables, and columns (3) and (4) are regression results with the inclusion of control variables. The t-statistics in parentheses are derived based on robust standard errors. Industry and year fixed effects are controlled for in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

Table 4
Firm-fixed effect.

	(1)	(2)
	Cash	Net Lev
ESG	0.0046**	−0.0159***
	(2.32)	(−5.26)
Bsize	0.0252**	−0.0392**
	(2.37)	(−2.50)
Size	−0.0110***	0.0827***
	(−3.58)	(16.13)
Roa	0.5258***	−1.0204***
	(15.16)	(−19.10)
Ownership	0.0496**	−0.0504
	(2.56)	(−1.56)
Firmage	−0.0059***	−0.0016*
	(−10.03)	(−1.84)
Indratio	−0.0110	−0.0077
	(−0.34)	(−0.16)
Tang	0.4393***	−0.3603***
	(20.17)	(−10.71)
Compensation	−0.0027	−0.0012
	(−1.12)	(−0.30)
Constant	0.0606	−0.6796***
	(0.80)	(−5.40)
Firm FE	Yes	Yes
Year FE	Yes	Yes
N	26,427	26,427
Adj-R2	0.223	0.243

Note: This table presents the OLS regression results of ESG performance on corporate risk-taking. The t-statistics in parentheses are derived based on robust standard errors. Firm and year fixed effects are controlled for in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

results in column (1) show that firms with larger size, higher return on assets, and a higher proportion of intangible assets are more likely to have better ESG performance. Columns (2) and (3) show the regression results after matching. The coefficient of Cash is positive and Net Lev is negative, which are both significant at the 1% level. The results indicate that after controlling for sample selection bias with PSM regression, the results are still consistent with the baseline result.

4.3.3. *Instrumental variable regression*

We choose the lagged term of the ESG score as the instrumental variable for two-step least squares regression to alleviate potential endogeneity. Since ESG performance is generally continuous, ESG performance in the previous period generally has a positive correlation with ESG performance in the current period but has less impact on the risk-taking choice of the firm in the current period. We understand that this choice might not be a perfect instrument variable, but it could help to verify our baseline results to some extent. The empirical results are shown in Table 6. The first-stage regressions of column (1) show a positive correlation between the current ESG score and the previous period at the 1% significance level. The second-stage regressions of columns (2) and (3) show a negative correlation between ESG and the level of corporate risk-taking at the 1% significance level, which is consistent with our previous finding.

4.3.4. *Alternative risk-taking measure*

To further check the robustness of the result with potential

Table 5
PSM regression.

	(1)	(2)	(3)
	ESG	Cash	Net Lev
ESG		0.0091***	−0.0223***
		(2.80)	(−4.35)
Bsize	0.1657	0.0202**	−0.0524***
	(1.44)	(2.32)	(−4.23)
Size	0.4393***	−0.0284***	0.0871***
	(24.48)	(−19.24)	(43.05)
Roa	21.2973***	0.6724***	−1.7621***
	(45.84)	(16.08)	(−29.60)
Ownership	0.7100***	0.0472***	−0.0709***
	(5.76)	(4.93)	(−5.16)
Firmage	0.0074**	−0.0016***	0.0032***
	(2.17)	(−6.10)	(8.45)
Indratio	0.6266	0.1025***	−0.1510***
	(1.47)	(3.20)	(−3.27)
Tang	0.9086***	0.2256***	−0.1241***
	(3.96)	(15.15)	(−5.29)
Compensation	0.4428***	0.0131***	−0.0130***
	(15.55)	(5.69)	(−4.03)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	26,427	7198	7198
r2_p	0.203		
Adj-R2		0.260	0.532

Note: Column (1) shows the regression results of the logit model, and columns (2) and (3) show the OLS regression results of the matched samples. The t-statistics in parentheses are calculated based on robust standard errors. Industry and year fixed effects are controlled for in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

Table 6
2SLS regression.

	(1)	(2)	(3)
	ESG	Cash	Net Lev
ESG		0.0319***	−0.0867***
		(8.62)	(−15.24)
Bsize	0.0234	0.0159***	−0.0186**
	(1.44)	(3.19)	(−2.42)
Size	0.0470***	−0.0260***	0.0907***
	(18.34)	(−31.23)	(71.01)
Roa	4.2066***	0.5241***	−1.3925***
	(61.62)	(17.73)	(−30.67)
Ownership	0.0302*	0.0385***	−0.0366***
	(1.71)	(7.08)	(−4.38)
Firmage	0.0016***	−0.0006***	0.0020***
	(3.46)	(−3.84)	(8.90)
Indratio	0.0785	0.0709***	−0.0872***
	(1.31)	(3.85)	(−3.08)
Tang	−0.0240	0.1783***	−0.0502***
	(−0.80)	(19.27)	(−3.53)
Compensation	0.0246***	0.0089***	−0.0050***
	(6.10)	(7.06)	(−2.60)
L.ESG	0.4636***		
	(82.87)		
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
N	21,306	21,306	21,306
Adj-R2	0.523	0.212	0.447

Note: This table presents the results of a two-stage least squares (2SLS) regression of the impact of ESG performance on corporate risk-taking. Control variables are included in each regression. Industry and year fixed effects are controlled for in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

measurement bias, we use four alternative risk-taking measures as defined in section 3. The results of the empirical regressions are shown in Table 7. The results of columns (1), (2) and (3) show that the coefficients of ESG are significantly negative at the 1% significance level, indicating that ESG performance significantly reduces the risk-taking level of the firm. In the regression results in column (4), the coefficient of ESG is positive at the 1% significance level, which indicates that ESG performance contributes to a significant increase in the distance from the default threshold and reduces the risk-taking level of the firm. Therefore, the regression results of the above four alternative risk-taking proxy variables are consistent with the empirical results of the main regression in this paper.

Table 7
Alternative measures of risk-taking.

	(1)	(2)	(3)	(4)
	Risk	Risk2	Risk3	DD Merton
ESG	−0.0048***	−0.0114***	−0.0156***	0.0476***
	(−5.41)	(−5.31)	(−3.83)	(8.25)
Bsize	−0.0061**	−0.0153**	−0.0021	0.0957***
	(−2.19)	(−2.31)	(−0.16)	(5.67)
Size	−0.0013***	−0.0033***	0.0055**	−0.0690***
	(−2.96)	(−3.12)	(2.29)	(−24.03)
Roa	0.0208	0.0680*	0.5461***	1.9712***
	(1.37)	(1.85)	(8.26)	(25.43)
Ownership	−0.0188***	−0.0442***	0.0674***	−0.0209
	(−6.48)	(−6.35)	(4.20)	(−1.10)
Firmage	0.0001	0.0004	−0.0010**	0.0054***
	(1.45)	(1.60)	(−2.13)	(10.42)
Indratio	0.0064	0.0150	0.1614***	0.2950***
	(0.64)	(0.63)	(3.08)	(4.70)
Tang	−0.0917***	−0.2163***	−0.2012***	0.1161***
	(−10.79)	(−10.79)	(−6.29)	(3.60)
Compensation	−0.0042***	−0.0101***	0.0035	0.0038
	(−5.72)	(−5.75)	(0.96)	(0.87)
Constant	0.2592***	0.6249***	0.2871***	2.8818***
	(18.33)	(18.64)	(4.70)	(36.27)
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
N	10,795	10,795	10,795	26,395
Adj-R2	0.118	0.121	0.061	0.295

Note: This table presents the results of OLS regressions replacing corporate risk-taking with different indicators. Control variables are included in each regression. The t-statistics in parentheses are derived based on robust standard errors. Industry and year fixed effects are controlled for in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

5. Mechanism analysis: Cross-sectional heterogeneity

5.1. Information asymmetry

Information transparency is higher in firms with lower financial constraints or firms being audited by Big 4 auditing firms. A lower level of corporate financing constraint also tends to affect firms' investment decisions. The lower the level of corporate financing constraint is, the more likely firms are to invest in long-term profitable projects. Therefore, firms with lower corporate financing constraints value the future value of the firm more, which is conducive to the investment opportunity of maximizing firm value. Firms will not give up future value investment due to the lack of capital, and the risk-taking level of the firm will be higher. Moreover, firms with higher financing constraints focus more on the reputation and trust of external stakeholders brought by ESG to release financial constraints, which can bring more financing advantages to firms and provide invisible guarantees for their external financing. For such advantages, firms will be reluctant to engage in risky investment behaviors to ruin their reputation.

The regression results are reported in Table 8. The *WW* indicator (Whited & Wu, 2006) is selected as the grouping variable to measure the level of firm financing constraints. The median value of financing constraints in each year and industry is used to divide the sample into low and high financial constraint groups. The regression results in columns

Table 8
Heterogeneity of financial constraints & Audit.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High-WW Cash	Low-WW Cash	High-WW Net Lev	Low-WW Net Lev	Big4 Cash	Non-Big4 Cash	Big4 Net Lev	Non-Big4 Net Lev
ESG	0.0190***	0.0015	−0.0508***	−0.0058*	0.0064	0.0151***	−0.0092	−0.0343***
	(7.29)	(0.64)	(−10.96)	(−1.65)	(0.89)	(8.84)	(−0.91)	(−11.79)
Bsize	0.0163**	0.0071	0.0256**	−0.0289***	−0.0202	0.0183***	−0.0117	−0.0118
	(2.00)	(1.17)	(1.98)	(−3.22)	(−1.37)	(3.69)	(−0.60)	(−1.57)
Size	−0.0372***	−0.0178***	0.1069***	0.0680***	−0.0219***	−0.0291***	0.0668***	0.0929***
	(−20.89)	(−14.44)	(36.52)	(38.23)	(−8.47)	(−34.43)	(19.28)	(74.67)
Roa	0.7872***	0.7091***	−1.5892***	−2.0902***	0.5758***	0.7002***	−1.9064***	−1.7461***
	(17.82)	(20.83)	(−22.22)	(−43.76)	(6.25)	(27.28)	(−17.25)	(−44.82)
Ownership	0.0331***	0.0373***	−0.0404***	−0.0340***	−0.0207	0.0424***	−0.0326	−0.0485***
	(3.70)	(5.34)	(−2.96)	(−3.26)	(−1.13)	(7.69)	(−1.24)	(−5.89)
Firmage	−0.0014***	−0.0014***	0.0036***	0.0035***	−0.0009	−0.0013***	0.0033***	0.0030***
	(−6.10)	(−7.11)	(10.07)	(11.78)	(−1.52)	(−8.81)	(4.60)	(13.57)
Indratio	0.0642**	0.0443*	−0.0171	−0.1015***	−0.0484	0.0788***	0.0333	−0.0792***
	(2.15)	(1.96)	(−0.37)	(−3.01)	(−0.85)	(4.27)	(0.44)	(−2.86)
Tang	0.2488***	0.1466***	−0.1468***	0.0076	0.1789***	0.2180***	−0.1110***	−0.0830***
	(19.84)	(15.04)	(−6.52)	(0.46)	(6.88)	(28.42)	(−2.78)	(−6.20)
Compensation	0.0115***	0.0109***	−0.0229***	−0.0074***	0.0133***	0.0107***	−0.0006	−0.0134***
	(5.54)	(6.99)	(−7.25)	(−3.24)	(3.41)	(8.43)	(−0.11)	(−6.98)
Constant	0.4821***	0.2164***	−1.1336***	−0.6314***	0.2460***	0.3377***	−0.4358***	−0.9711***
	(10.62)	(6.91)	(−15.16)	(−13.72)	(3.34)	(14.47)	(−4.08)	(−26.91)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	11,059	11,147	11,059	11,147	1582	24,845	1582	24,845
Adj-R2	0.282	0.254	0.410	0.494	0.245	0.252	0.564	0.447

Note: This table shows the regression results of subsamples classified by the level of financial constraints and whether the firm is audited by a Big 4 auditing firm. Columns (1) to (4) show the regression results of the financially constrained groups, and columns (5) to (8) show the regression results for the Big 4 audit groups. Control variables are included in each regression. The t-statistics in parentheses are calculated based on robust standard errors. Industry and year fixed effects are controlled in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

(1) and (3) show that ESG performance has a significant effect on reducing the risk-taking level of firms in the high financial constraint group, and the coefficients are both insignificant in columns (2) and (4). Similarly, the coefficient of ESG is only significant in firms audited by non-Big 4 auditors in columns (6) and (8). This empirical result verifies hypothesis H2 of this paper. The marginal impact of ESG on corporate risk-taking is more pronounced in firms with lower information transparency.

Table 9
Heterogeneity of institutional shareholdings (INI) & Managers Perks (MP).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High-INI Cash	Low- INI Cash	High- INI Net Lev	Low INI Net Lev	High- MP Cash	Low- MP Cash	High- MP Net Lev	Low- MP Net Lev
ESG	0.0104***	0.0217***	−0.0277***	−0.0460***	0.0148***	0.0132***	−0.0359***	−0.0305***
	(4.95)	(8.19)	(−8.02)	(−10.19)	(5.72)	(6.29)	(−8.33)	(−8.71)
Bsize	0.0199***	0.0112	−0.0320***	0.0121	0.0147**	0.0190***	0.0104	−0.0321***
	(3.31)	(1.52)	(−3.46)	(1.12)	(1.96)	(3.22)	(0.94)	(−3.66)
Size	−0.0222***	−0.0359***	0.0810***	0.0998***	−0.0278***	−0.0227***	0.0836***	0.0767***
	(−21.63)	(−28.97)	(53.61)	(54.95)	(−21.23)	(−22.39)	(43.75)	(52.07)
Roa	0.7474***	0.6260***	−1.9431***	−1.5145***	0.6823***	0.7331***	−1.5917***	−1.9919***
	(23.29)	(16.22)	(−42.18)	(−25.38)	(17.83)	(22.90)	(−27.26)	(−42.95)
Ownership	0.0186**	0.0548***	−0.0294***	−0.0707***	0.0499***	0.0362***	−0.0504***	−0.0625***
	(2.55)	(6.79)	(−2.67)	(−6.00)	(6.19)	(5.33)	(−4.24)	(−6.22)
Firmage	−0.0009***	−0.0014***	0.0029***	0.0029***	−0.0017***	−0.0008***	0.0035***	0.0027***
	(−4.63)	(−7.09)	(9.49)	(9.80)	(−7.93)	(−4.29)	(10.78)	(9.89)
Indratio	0.0667***	0.0558**	−0.1286***	0.0071	0.0746***	0.0486**	−0.0532	−0.0905***
	(2.93)	(2.09)	(−3.77)	(0.18)	(2.69)	(2.24)	(−1.32)	(−2.75)
Tang	0.1978***	0.2219***	−0.0737***	−0.0892***	0.2467***	0.2049***	−0.1860***	−0.0425**
	(19.61)	(20.82)	(−4.20)	(−4.84)	(24.22)	(18.78)	(−10.59)	(−2.32)
Compensation	0.0076***	0.0155***	−0.0084***	−0.0200***	0.0109***	0.0071***	−0.0150***	−0.0022
	(4.81)	(8.37)	(−3.55)	(−7.30)	(5.72)	(4.63)	(−5.29)	(−0.98)
Constant	0.2334***	0.4498***	−0.6754***	−1.1346***	0.3141***	0.2464***	−0.7215***	−0.7456***
	(8.30)	(13.00)	(−15.53)	(−21.49)	(9.16)	(8.96)	(−13.83)	(−17.52)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	13,171	13,256	13,171	13,256	13,211	13,215	13,211	13,215
Adj-R2	0.225	0.283	0.463	0.450	0.252	0.249	0.403	0.494
Diff	−0.0113***		0.0183***		0.0016***		−0.0054***	

Note: This table shows the regression results of subsamples classified by the level of institutional ownership and manager perks. Columns (1) to (4) show the regression results for institutional ownership groups, and columns (5) to (8) show the regression results for manager perks groups. Control variables are included in each regression. The t-statistics in parentheses are calculated based on robust standard errors. Industry and year fixed effects are controlled in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

5.2. Corporate governance

We argue that the effect of ESG on corporate risk-taking will be different based on corporate governance characteristics. Institutional investors, as professional investors in the market, have more professional analysis skills and information advantages. Institutional investors use their professional ability to obtain more information to reduce the information asymmetry of the firm. Therefore, accounting information quality will be better for firms with higher ratio of institutional investors (Bushee, 1998; He, Qin, Liu, & Wu, 2022). In addition, more institutional investor monitoring will reduce the potential self-intereste behavior at the expense of shareholders (An & Zhang, 2013; Xu, Li, Yuan, & Chan, 2014). Institutional investors could provide firms with better investment suggestions with their shareholding advantages and information advantages. Therefore, firms with a higher percentage of institutional investors tend to have lower agency costs and make investment decisions based on the future value of the firm. Meanwhile, the lower level of perks of managers also implies a lower agency cost.

The results of the cross-sectional heterogeneity analysis in corporate governance are shown in Table 9. The median value of institutional investors' proportion and managers' perks in each year and industry are used to divide the sample into better and weaker corporate governance groups, respectively. The regression results in columns (1) to (4) show that ESG in reducing the risk-taking level of firms is more pronounced in firms with lower institutional ownership, and columns (5) to (8) show that ESG in reducing the risk-taking level of firms is more pronounced in firms with higher manager perks. Overall, these empirical results verify hypothesis H3 of this paper. The marginal impact of ESG on corporate risk-taking is more pronounced in firms with weaker corporate governance.

5.3. External monitoring

External monitoring is crucial to firm development. Analysts, as professional investors and information interpreters, play an important role in collecting internal and external information about firms and evaluating their future profitability. Analysts can use their professional knowledge to filter out effective information from the complex information environment, making their own professional analysis and judgment on the ESG behavior of firms, thus reducing the information asymmetry of firms. Analyst coverage and research repost alleviate the information asymmetry of firms, thus reducing the cost of financing and alleviating financing constraints (Cheng, Ioannou, & Serafeim, 2014; Dhaliwal, Li, Tsang, & Yang, 2011; He, Feng, & Hao, 2022). Therefore, the more attention a firm receives from external financial analysts, the stronger external monitoring power it has, thus weakening the marginal impact of ESG.

The results of the cross-sectional heterogeneity analysis in external monitoring are shown in Table 10. The median value of analyst coverage and analysts' reports in each year and industry are used to divide the sample into low and high external monitoring groups, respectively. The regression results in columns (2) and (4) show that ESG only has a significant effect on reducing the risk-taking level of firms in the lower analyst coverage group, and columns (6) and (8) show that ESG only has a significant effect on reducing the risk-taking level in firms with fewer analysts' reports. These empirical results verify hypothesis H3 of this paper. The marginal impact of ESG on corporate risk-taking is more pronounced in firms with lower external monitoring power

6. Conclusions

This paper explores the impact of corporate ESG performance on its risk-taking level. Corporate risk-taking is measured by joint cash holding and net leverage level. We find that ESG performance significantly reduces corporate risk-taking, resulting in a more stable operating environment for firms. Managers consider ESG as a safe tool for investment at the loss of firm long-term value. As a result, the marginal impact of ESG performance on corporate risk-taking is more pronounced in firms with lower transparency, weaker corporate governance and less external monitoring pressure. Our results provide important empirical evidence

Table 10
Heterogeneity of analyst coverage and research reports.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	High-Analyst Cash	Low-Analyst Cash	High-Analyst Net Lev	Low-Analyst Net Lev	High-Report Cash	Low-Report Cash	High-Report Net Lev	Low-Report Net Lev
ESG	−0.0009	0.0117***	−0.0005	−0.0299***	−0.0009	0.0117***	−0.0006	−0.0298***
	(−0.31)	(4.49)	(−0.12)	(−6.89)	(−0.31)	(4.49)	(−0.13)	(−6.88)
Bsize	0.0027	0.0128	−0.0284***	−0.0050	0.0027	0.0128	−0.0285***	−0.0050
	(0.35)	(1.60)	(−2.69)	(−0.43)	(0.36)	(1.60)	(−2.70)	(−0.42)
Size	−0.0273***	−0.0319***	0.0830***	0.0964***	−0.0273***	−0.0319***	0.0831***	0.0964***
	(−19.59)	(−23.33)	(43.90)	(47.57)	(−19.60)	(−23.33)	(43.92)	(47.56)
Roa	0.6144***	0.6919***	−1.7857***	−1.7898***	0.6141***	0.6920***	−1.7848***	−1.7902***
	(16.12)	(14.95)	(−34.51)	(−25.73)	(16.11)	(14.96)	(−34.50)	(−25.73)
Ownership	0.0362***	0.0307***	−0.0328***	−0.0342***	0.0362***	0.0308***	−0.0327***	−0.0344***
	(4.12)	(3.57)	(−2.66)	(−2.68)	(4.12)	(3.58)	(−2.65)	(−2.69)
Firmage	−0.0015***	−0.0011***	0.0029***	0.0026***	−0.0015***	−0.0011***	0.0029***	0.0026***
	(−6.50)	(−4.56)	(8.89)	(7.44)	(−6.51)	(−4.54)	(8.93)	(7.42)
Indratio	0.0240	0.1174***	−0.0928**	−0.1240***	0.0239	0.1175***	−0.0925**	−0.1244***
	(0.83)	(4.07)	(−2.33)	(−2.88)	(0.83)	(4.07)	(−2.32)	(−2.89)
Tang	0.2155***	0.2239***	−0.0788***	−0.0716***	0.2156***	0.2238***	−0.0792***	−0.0714***
	(18.48)	(18.40)	(−4.24)	(−3.46)	(18.49)	(18.39)	(−4.26)	(−3.45)
Compensation	0.0093***	0.0099***	−0.0036	−0.0104***	0.0093***	0.0099***	−0.0036	−0.0104***
	(4.76)	(4.96)	(−1.33)	(−3.45)	(4.76)	(4.96)	(−1.32)	(−3.46)
Constant	0.4424***	0.4059***	−0.9859***	−1.1116***	0.4426***	0.4058***	−0.9866***	−1.1112***
	(11.88)	(10.78)	(−19.01)	(−18.96)	(1)	(2)	(3)	(4)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	9767	10,257	9767	10,257	9768	10,256	9768	10,256
Adj-R2	0.278	0.257	0.523	0.465	0.278	0.257	0.523	0.465

Note: This table shows the regression results of subsamples classified by analyst coverage and reports. Columns (1) to (4) show the regression results of analyst coverage groups, and columns (5) to (8) show the regression results for analyst report groups. Control variables are included in each regression. The t-statistics in parentheses are calculated based on robust standard errors. Industry and year fixed effects are controlled in each regression. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively. All the variable definitions are presented in Table 1.

of the impact of ESG on a firm's operating environment in a transition economy in China, which also generates implications for countries worldwide.

Based on the above empirical findings, we argue that the impact of ESG on the risky choices of firms may be detrimental to the accumulation of social capital at the aggregate level, and the conservative choices of enterprises may inhibit innovative capital expenditures. Therefore, we should encourage firms to make innovative investments while evaluating ESG performance. To provide better positive ESG feedback, we should strengthen the level of external supervision in the capital market and optimize the internal governance of firms to avoid adverse selection and moral hazard caused by information asymmetry, which is conducive to the growth of long-term innovation ability and the formation of core competitiveness. We should improve the disclosure mechanism of social responsibility, enhance the disclosure effect, and promote the risk-taking choice of firms to be more consistent with the future value of enterprises.

Declaration of Competing Interest

None.

Data availability

Data will be made available on request.

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References

Acemoglu, D., & Zilibotti, F. (1997). Was Prometheus unbound by chance? Risk, diversification, and growth[J]. *Journal of Political Economy, 105*(4), 709–751.
Acharya, V. V., Amihud, Y., & Litov, L. (2011). Creditor rights and corporate risk-taking [J]. *Journal of Financial Economics, 102*(1), 150–166.
Akbar, S., Kharabsheh, B., Poletti-Hughes, J., & Shah, S. Z. A. (2017). Board structure and corporate risk taking in the UK financial sector[J]. *International Review of Financial Analysis, 50*, 101–110.
Almeida, H., & Campello, M. (2007). Financial constraints, asset tangibility, and corporate investment. *The Review of Financial Studies, 20*(5), 1429–1460.
An, H., & Zhang, T. (2013). Stock price synchronicity, crash risk, and institutional investors. *Journal of Corporate Finance, 21*, 1–15.
Anantharaman, D., & Lee, Y. G. (2014). Managerial risk taking incentives and corporate pension policy[J]. *Journal of Financial Economics, 111*(2), 328–351.
Baker, M., & Wurgler, J. (2013). Behavioral corporate finance: An updated survey[M]. In *, 2. Handbook of the Economics of Finance* (pp. 357–424). Elsevier.
Bargeron, L. L., Lehn, K. M., & Zutter, C. J. (2010). Sarbanes-Oxley and corporate risk-taking[J]. *Journal of Accounting and Economics, 49*(1–2), 34–52.
Baumol, W. J., & Strom, R. J. (2007). Entrepreneurship and economic growth[J]. *Strategic Entrepreneurship Journal, 1*(3–4), 233–237.
Bernile, G., Bhagwat, V., & Rau, P. R. (2017). What does not kill you will only make you more risk-loving: Early-life disasters and CEO behavior[J]. *The Journal of Finance, 72*(1), 167–206.
Borghesi, R., Houston, J. F., & Naranjo, A. (2014). Corporate socially responsible investments: CEO altruism, reputation, and shareholder interests[J]. *Journal of Corporate Finance, 26*, 164–181.
Boubakri, N., Cosset, J. C., & Saffar, W. (2013). The role of state and foreign owners in corporate risk-taking: Evidence from privatization[J]. *Journal of Financial Economics, 108*(3), 641–658.
Bushee, B. J. (1998). The influence of institutional investors on myopic R&D investment behavior[J]. *Accounting Review*, 305–333.
Carpenter, M. A., Pollock, T. G., & Leary, M. M. (2003). Testing a model of reasoned risk-taking: Governance, the experience of principals and agents, and global strategy in high-technology IPO firms[J]. *Strategic Management Journal, 24*(9), 803–820.
Cheng, B., Ioannou, I., & Serafeim, G. (2014). Corporate social responsibility and access to finance[J]. *Strategic Management Journal, 35*(1), 1–23.
Coles, J. L., Daniel, N. D., & Naveen, L. (2006). Managerial incentives and risk-taking[J]. *Journal of Financial Economics, 79*(2), 431–468.
Daron, A., & Zilibotti, F. (1997). Was Prometheus unbound by chance? Risk, diversification, and growth[J]. *Journal of Political Economy, 105*(4), 709–751.
Dhaliwal, D. S., Li, O. Z., Tsang, A., & Yang, Y. G. (2011). Voluntary nonfinancial disclosure and the cost of equity capital: The initiation of corporate social responsibility reporting. *The Accounting Review, 86*(1), 59–100.
Dunbar, C. G., Li, Z. F., & Shi, Y. (2020). Corporate social responsibility and CEO risk-taking incentives[J]. *Journal of Corporate Finance, 64*(101714).
Eisenmann, T. R. (2002). The effects of CEO equity ownership and firm diversification on risk taking[J]. *Strategic Management Journal, 23*(6), 513–534.
Eling, M., & Marek, S. D. (2014). Corporate governance and risk taking: Evidence from the UK and German insurance markets[J]. *The Journal of Risk and Insurance, 81*(3), 653–682.
Faccio, M., Marchica, M. T., & Mura, R. (2011). Large shareholder diversification and corporate risk-taking[J]. *The Review of Financial Studies, 24*(11), 3601–3641.
Faccio, M., Marchica, M. T., & Mura, R. (2016). CEO gender, corporate risk-taking, and the efficiency of capital allocation[J]. *Journal of Corporate Finance, 39*, 193–209.
Fazzari, S. M., & Athey, M. J. (1987). Asymmetric information, financing constraints, and investment[J]. *The Review of Economics and Statistics*, 481–487.
Godfrey, P. C., Hatch, N. A., & Hansen, J. M. (2005). Corporate social responsibility: Theory and evidence[C]. In *, 16. Proceedings of the International Association for Business and Society* (pp. 112–117).
Godfrey, P. C., Merrill, C. B., & Hansen, J. M. (2009). The relationship between corporate social responsibility and shareholder value: An empirical test of the risk management hypothesis[J]. *Strategic Management Journal, 30*(4), 425–445.
Gopalan, R., Gormley, T. A., & Kalda, A. (2021). It's not so bad: Director bankruptcy experience and corporate risk-taking[J]. *Journal of Financial Economics, 142*(1), 261–292.
Groza, M. D., Pronschinske, M. R., & Walker, M. (2011). Perceived organizational motives and consumer responses to proactive and reactive CSR[J]. *Journal of Business Ethics, 102*(4), 639–652.
Guariglia, A. (2008). Internal financial constraints, external financial constraints, and investment choice: Evidence from a panel of UK firms[J]. *Journal of Banking & Finance, 32*(9), 1795–1809.
Habib, A., & Hasan, M. M. (2017). Firm life cycle, corporate risk-taking and investor sentiment[J]. *Accounting and Finance, 57*(2), 465–497.
Hao, J, & He, F. (2022). Corporate social responsibility (CSR) performance and green innovation: Evidence from China. *Finance Research Letters, 48*, 102889.
He, F., Du, H., & Yu, B. (2022). Corporate ESG performance and manager misconduct: Evidence from China. *International Review of Financial Analysis, 82*, Article 102201.
He, F, Feng, Y, & Hao, J. (2022). Information disclosure source, investors' searching and stock price crash risk. *Economics Letters, 210*, 110202.
He, F, Qin, S, Liu, Y, & Wu, J. (2022). CSR and idiosyncratic risk: Evidence from ESG information disclosure. *Finance Research Letters, 49*, 102936.
Hirshleifer, D., & Thakor, A. V. (1992). Managerial conservatism, project choice, and debt. *The Review of Financial Studies, 5*(3), 437–470.
Ho, H., Kim, N., & Reza, S. (2022). CSR and CEO pay: Does CEO reputation matter?[J]. *Journal of Business Research, 149*, 1034–1049.
Hong, B., Li, Z., & Minor, D. (2016). Corporate governance and executive compensation for corporate social responsibility[J]. *Journal of Business Ethics, 136*(1), 199–213.
Houston, J. F., & James, C. M. (2001). Do relationships have limits? Banking relationships, financial constraints, and investment[J]. *The Journal of Business, 74*(3), 347–374.
Ioannou, I., & Serafeim, G. (2015). The impact of corporate social responsibility on investment recommendations: Analysts' perceptions and shifting institutional logics. *Strategic Management Journal, 36*(7), 1053–1081.
Jiang, F., Jiang, Z., Kim, K. A., et al. (2015). Family-firm risk-taking: Does religion matter?[J]. *Journal of Corporate Finance, 33*, 260–278.
Jiraporn, P., Chatjuthamard, P., Tong, S., & Kim, Y. S. (2015). Does corporate governance influence corporate risk-taking? Evidence from the institutional shareholders services (ISS)[J]. *Finance Research Letters, 13*, 105–112.
Jo, H., & Na, H. (2012). Does CSR reduce firm risk? Evidence from controversial industry sectors[J]. *Journal of Business Ethics, 110*(4), 441–456.
John, K., Litov, L., & Yeung, B. (2008). Corporate governance and risk-taking[J]. *The Journal of Finance, 63*(4), 1679–1728.
Kahneman, D., & Tversky, A. (1979). *On the interpretation of intuitive probability: A reply to Jonathan Cohen[J]*.
Kim, K., Patro, S., & Pereira, R. (2017). Option incentives, leverage, and risk-taking[J]. *Journal of Corporate Finance, 43*, 1–18.
Li, D. (2011). Financial constraints, R&D investment, and stock returns[J]. *The Review of Financial Studies, 24*(9), 2974–3007.
Li, J., & Tang, Y. I. (2010). CEO hubris and firm risk taking in China: The moderating role of managerial discretion. *Academy of Management Journal, 53*(1), 45–68.
Li, K., Griffin, D., Yue, H., & Zhao, L. (2013). How does culture influence corporate risk-taking?[J]. *Journal of Corporate Finance, 23*, 1–22.
Lim, E. N. K., & McCann, B. T. (2014). Performance feedback and firm risk taking: The moderating effects of CEO and outside director stock options[J]. *Organization Science, 25*(1), 262–282.
Lins, K. V., Servaes, H., & Tamayo, A. (2017). Social capital, trust, and firm performance: The value of corporate social responsibility during the financial crisis[J]. The. *Journal of Finance, 72*(4), 1785–1824.
Low, A. (2009). Managerial risk-taking behavior and equity-based compensation[J]. *Journal of Financial Economics, 92*(3), 470–490.
Luo, C., Wei, D., & He, F. (2023). Corporate ESG performance and trade credit financing – Evidence from China[J]. *International Review of Economics & Finance, 85*, 337–351.
MacCrimmon, K. R., & Wehrung, D. A. (1990). Characteristics of risk taking executives [J]. *Management Science, 36*(4), 422–435.

Oikonomou, I., Brooks, C., & Pavelin, S. (2012). The impact of corporate social performance on financial risk and utility: A longitudinal analysis[J]. *Financial Management, 41*(2), 483–515.
Orlitzky, M., & Benjamin, J. D. (2001). Corporate social performance and firm risk: A meta-analytic review[J]. *Business & Society, 40*(4), 369–396.
Phan, H. V., & Hegde, S. P. (2013). Corporate governance and risk taking in pension plans: Evidence from defined benefit asset allocations[J]. *Journal of Financial and Quantitative Analysis, 48*(3), 919–946.
Porter, M. E., & Kramer, M. R. (2006). The link between competitive advantage and corporate social responsibility[J]. *Harvard Business Review, 84*(12), 78–92.
Quigley, T. J., Hambrick, D. C., Misangyi, V. F., et al. (2019). CEO selection as risk-taking: A new vantage on the debate about the consequences of insiders versus outsiders[J]. *Strategic Management Journal, 40*(9), 1453–1470.
Santosh, K., Andrew, M., Suman, N., & Chandra, T. (2018). Corporate governance reform and risk-taking: Evidence from a quasi-natural experiment in an emerging market [J]. *Journal of Corporate Finance, 61*, 103996.
Servaes, H., & Tamayo, A. (2013). The impact of corporate social responsibility on firm value: The role of customer awareness[J]. *Management Science, 59*(5), 1045–1061.
Sila, V., Gonzalez, A., & Hagendorff, J. (2016). Women on board: Does boardroom gender diversity affect firm risk?[J]. *Journal of Corporate Finance, 36*, 26–53.
Su, W., & Lee, C. Y. (2013). Effects of corporate governance on risk taking in Taiwanese family firms during institutional reform[J]. *Asia Pacific Journal of Management, 30* (3), 809–828.
Suharyono, S. (2019). The effect of accountability, transparency, and supervision on budget performance by using the concept of value for money in regional business enterprises (bumd) of Riau province[J]. *International Journal of Public Finance, 4*(2), 236–249.
Wang, C. J. (2012). Board size and firm risk-taking[J]. *Review of Quantitative Finance and Accounting, 38*(4), 519–542.
Whited, T. M., & Wu, G. (2006). Financial constraints risk. *The Review of Financial Studies, 19*(2), 531–559.
Xu, N., Li, X., Yuan, Q., & Chan, K. C. (2014). Excess perks and stock price crash risk: Evidence from China. *Journal of Corporate Finance, 25*, 419–434.